

Homework (Habanero)**Q1.** Evaluate the expression: $3 \cdot 2 + 12 \div 4$

- (A) 9
(B) 10
(C) 11
(D) 12
(E) 13

Find the number of bacteria after 2 hours.

- (A) 15
(B) 17
(C) 19
(D) 23
(E) 25

Q2. A patient needs 2 pills every 4 hours. If x is the number of hours, how many pills will the patient need in x hours?

- (A) $\frac{x}{2}$
(B) $\frac{x}{4}$
(C) $\frac{2x}{4}$
(D) $\frac{x}{8}$
(E) $\frac{2x}{8}$

Q6. Simplify the expression: $4(3x - 2) - 2(2x + 1)$

- (A) $8x - 10$
(B) $8x - 12$
(C) $10x - 8$
(D) $12x - 6$
(E) $10x - 10$

Q3. The expression $3(2x - 1) + 2$ simplifies to

- (A) $6x - 1$
(B) $6x + 1$
(C) $6x - 3$
(D) $6x + 5$
(E) $6x - 5$

Q7. A patient's temperature is given by the expression $98.6 + 0.5x - 2$, where x is the number of hours since the patient was admitted. What is the patient's temperature after 4 hours?

- (A) 100.0
(B) 100.2
(C) 101.0
(D) 101.4
(E) 102.0

Q4. Evaluate the expression: $(8 - 3)^2 + 2 \cdot 4$

- (A) 45
(B) 50
(C) 55
(D) 65
(E) 70

Q8. The expression $(2x + 1)^2 - 4$ simplifies to

- (A) $4x^2 - 3$
(B) $4x^2 + 1$
(C) $4x^2 + 3$
(D) $4x^2 - 1$
(E) $4x^2 - 4$

Q5. The number of bacteria in a culture grows according to the expression $2x^2 + 5x + 1$, where x is the number of hours.**Open Ended Questions (FRQ)****Q9.** A doctor prescribes a medication that decays at a rate given by the expression $5 - 2x$, where x is the number of days since the medication was prescribed. Find the rate of decay after 3 days and show all work. Give your final answer as a numerical value.**Q10.** The cost of producing x units of a certain medication is given by the expression $2000 + 10x + 0.5x^2$. Find the cost of producing 20 units and show all work. Give your final answer as a numerical value.**Chef's Tips**

- Chef's Tips: Please ensure students have a calculator and a pencil for this test.
- Chef's Tips: The questions are designed to test the students' understanding of the topics and are meant to be challenging.
- Chef's Tips: Make sure students have enough space to show their work for the free response questions.